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Stat 444 Exam 1 Rev

3-16-22

1. a) Type I error in hypothesis testing
probability of rejecting the null hypothesis when it is true
 $PC(\text{reject } H_0 | H_0 \text{ true}) = \alpha$

b) Type II error in hypothesis testing
 $PC(\text{do not reject null} | \text{null false}) = \beta = 1 - \text{power}$

c) p-value or test

- probability of rejecting null hypothesis for a particular test, based on test statistic and compared to α
- $PC(Z < -t_{\alpha/2} | \text{or } PC(Z > t_{\alpha/2} | \text{or } PC(Z < -t_{\alpha/2} | \text{or } PC(Z > t_{\alpha/2} |$ (assuming standard normal, also test using right-tail 2-tail appropriate to $t, \chi^2, etc.$)
- probability that the result occurred by random chance
- as p-value $\downarrow$, statistical significance $\uparrow$

2.

Group	$\bar{x}$	s
1	5.617	1.277
2	5.550	1.237
Distance	0.067	0.207

a) $\frac{s_1^2}{s_2^2} F_L < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} F_U$ assuming $CI \rightarrow \alpha = 0.05$
 $d.f. = d.f. = 5$ use $\alpha/2 = 0.025$

$s_1^2 = 1.0307$ $s_2^2 = 1.4301$

$F_L = \frac{1}{F_{0.025, 5, 5}}$
 $= 1/7.15$

$F_U = F_{0.025, 5, 5}$
 $= 7.15$

$CI: (0.1489, 7.6202)$

Explanation: Lower ratio means variances are very similar, so both new and control treatments show similar consistency. High ratio means the new treatment group shows more variance than the control, which could mean it is less reliable/effective.

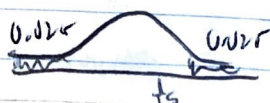
b) $\alpha = 0.15$ $H_0: \mu_1 = \mu_2$ $H_1: \mu_1 \neq \mu_2$

→ Chosen t-test for μ_1, μ_2 : independent samples, approximately normal, with equal variances
the $\rightarrow$ small sample size $\rightarrow$ square game $df = 6 + 6 - 2 = 10$

$ts = \frac{5.617 - 5.550}{s_p \sqrt{\frac{1}{6} + \frac{1}{6}}}$
 ≥ 0.0734

p-value: $PC(t > 0.0734)$ from the t_0 what reject null
TS does not fall in rejection region at $\alpha = 0.05$

Obs $\pm t_{0.025, 10} = \pm 2.228$



3. $n=14$ $\alpha=0.05$ $\mu_0=12$ $\mu_a=10$

a) power = probability of rejecting null when it is false or $P(\text{reject } H_0 | H_0 \text{ false})$

$$\beta(\mu_a) = P\left(Z \leq Z_{\alpha} - \frac{|\mu_0 - \mu_a|}{\sigma/\sqrt{n}}\right) = P\left(Z \leq 1.645 - \frac{|10-12|}{1.5/\sqrt{14}}\right) = .214$$

$Z_{0.05} = 1.645$ $1.5/\sqrt{14}$

$1 - .214 = .786 \approx \text{power of test}$

b)

$$n = \frac{\sigma^2 (Z_{\alpha} + Z_{\beta})^2}{\Delta^2}$$

$$\Delta = \mu_0 - \mu_a$$

$\Delta = 2$

$$\beta = 1 - \text{power} \rightarrow = 0.2$$

$$Z_{\alpha} \rightarrow Z_{0.05} = 1.645$$

$$Z_{\beta} \rightarrow Z_{0.2} = .4615$$

$$n = \frac{(1.5)^2 (1.645 + .4615)^2}{4} \approx 3.418 \rightarrow \boxed{4, \text{ choose } n=4}$$

4.

$$H_0: \sigma^2 \geq 1 \quad H_a: \sigma^2 < 1 \quad n=16, \quad s^2=0.41 \quad \alpha=0.05$$

$$\chi^2 = \frac{15(0.41)}{1} = 12.15$$

$$CV = \chi^2_{15, 0.95} = 7.261$$

left-tailed!

TS χ^2 CV, we can conclude that the scale is not reliable because the null hypothesis isn't rejected at $\alpha=0.05$. Not enough evidence to reject the null hypothesis

P-value: $P(\chi^2 < 12.15) \rightarrow (0.1 < P < 0.9)$, greater than $\alpha=0.05$, we do not reject the null hypothesis

5. $n_1=14$ $n_2=14$ two-sided, use $\alpha/2 = 0.025 = p$

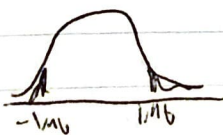
assumptions: Samples are independent, ANOVA can test because there were two separate sample groups (A and B). There is also the assumption that there are no tied ranks

$$Z = \frac{110 - 232.5}{24.109} = 4.66$$

$$\mu_T = \frac{n_1(n_1+n_2-1)}{2} = 232.5$$

$$\sigma_T = \sqrt{\frac{(225)}{12} (31)} = 24.1$$

RR:



4.66 falls into RR, so there is sufficient evidence to suggest a difference between the two types of wire.

g/m² 1: 40F g/m² 2: 60F

$\rightarrow H_0: \mu_1 = \mu_2$ vs $H_1: \mu_1 > \mu_2$ $\alpha = 0.05$

6. Track | d test: paired t-test

1 | 0.02

2 | 0.02

3 | 0.01

4 | 0.03

5 | 0.18

6 | 0

7 | 0.02

8 | 0.07

9 | -0.04

10 | 0.06

assumptions: non-independent samples (same track)

observe twice in each instance

approximately normal distribution

each element within the samples are

independent of each other (different tracks)

$D_0 = 0$

$\bar{d} = 0.037$

$S_d = 0.0537$

one-tailed, use $\alpha = 0.05$

$df = n - 1 = 10 - 1 = 9$

$\alpha = 0.05$ $t_{9, 0.05} = CV = 1.833$

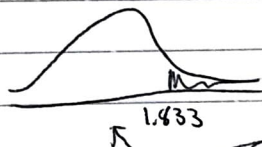
$TS = \frac{\bar{d} - D_0}{S_d / \sqrt{n}}$

$df = n - 1 = 9$

$S_d / \sqrt{n}$

$\rightarrow = \frac{0.037 - 0}{0.0537 / \sqrt{10}} = 0.1943$

RR:



TS.

Conclusion:

The test statistic does not

fall into the rejection region, so there

is not sufficient evidence to reject the null

at $\alpha = 0.05$

7. a) False, we must ~~test~~ know the null hypothesis is true.
i.e. it is given, α = chance of incorrectly rejecting the null

b) True, this means that there was no change between the speed limit and the expected speed of drivers

c) True, confidence is in the process, once the sample is taken there is no longer any randomness in the expression, so a randomised CI could not even be inaccurate

d) False, the F test is sensitive to the normality assumption

e) False, the F distribution is constructed as the ratio of two Chi-square random variables, (Normal random variables do not have degrees of freedom)

f) True